

GroupPaperA Essay

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ENGR 190W

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Editor:

Grading Categories:

1. Critical thinking & analysis:
2. Use of evidence/research:
3. Development & structure:
4. Language & style conventions:

Table of Contents

Part 1: Introduction/Background: The Why?

Part 2: Proposal Statement: The What?

Part 3: Work Product: The Evidence

Part 4: Target Customer: The Evidence

Part 5: Competition: The Evidence

Part 6: Method or Approach: The Evidence

Part 7: Risk Management: The Evidence

Part 8: Conclusion

Part 9: Reference

Part 1: Introduction/Background: The Why?

Atrial fibrillation (AFib) is an increasingly common heart condition that primarily affects older adults and individuals with certain risk factors. It is characterized by an irregular and often rapid heart rhythm, which leads to the chaotic beating of the atria (upper heart chambers). This irregular rhythm can result in blood clots forming in the heart, potentially leading to stroke, heart failure, and other serious heart-related complications. Although not always life-threatening on its own, AFib is a significant medical concern that demands careful management to prevent its severe consequences.

Many people living with AFib, especially elderly individuals, face the challenge of adhering to strict medication regimens, as these are often essential for preventing blood clots and managing the risk of stroke. However, forgetfulness, confusion over complex dosing schedules, and difficulty maintaining regular monitoring routines contribute to high rates of non-adherence. In fact, non-adherence is a major issue with notable impacts. A public health report published by the NIH states that 125,000 deaths are accounted for annually in the U.S. and adds approximately \$300 billion to healthcare costs. By 2050, with the global population aged 60 and older expected to double to 2.1 billion, the importance of addressing this issue becomes ever more pressing [1].

Studies highlight that around 86.3% of American adults aged 60–79 take at least one prescription medication, underscoring the scale of the adherence problem in older populations [2].

Part 2: Proposal Statement: The What?

To address the adherence and monitoring challenges faced by elderly individuals with AFib, we propose the development of a software application called HeartWatch designed specifically for AFib symptom monitoring and prescription adherence. HeartWatch will be compatible with popular wearable devices like Apple Watch, Fitbit, and Garmin, and it will leverage advanced IoT technology and AI-driven analytics to offer comprehensive health management.

HeartWatch will include features such as vital sign monitoring (tracking blood pressure, heart rate, and blood oxygen levels), motion sensor data analysis to confirm medication intake patterns, and real-time alerts to prompt users to take their medication or seek medical attention if abnormal vitals are detected. By synthesizing data from multiple health indicators and employing machine learning algorithms, HeartWatch can provide personalized feedback, helping users proactively manage their AFib. This technology solution thus addresses both the adherence issue and the need for continuous monitoring, supporting AFib patients in achieving improved health outcomes and reducing their risk of serious complications.

Part 3: Work Product: The Evidence

HeartWatch is a smart watch application that leverages machine learning (ML) algorithms to recognize abnormal symptoms associated with Afib and notify the user of any potential medical issues or emergencies. Secondly, it will use the same algorithmic structure to monitor hand and wrist motions to detect possible lapses in prescription use.

Modern smart watches possess capabilities that can be leveraged to track and analyze vital signs used in the detection and maintenance of Afib symptoms. The four leading devices by

global market share [3] all contain optical sensors called photoplethysmographs (PPG) and inertial measurement units (IMU) that contain accelerometers and gyroscopes. PPG sensors use green wavelength light emitting diodes (LED) and sensitive photodiodes to measure blood volume beneath the skin and these measurements are used to compute heart rate, blood pressure, and blood oxygen saturation. The PPG data which is cleaned and refined as a heartbeat waveform looks like the following:

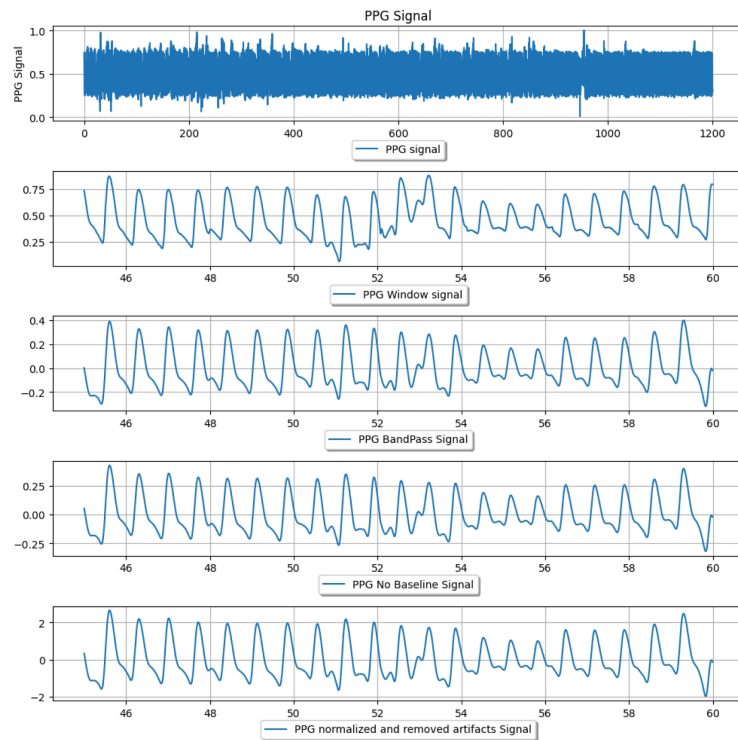


Figure 1 - Raw PPG data refined and normalized to be interpreted by our ML stack [4].

IMU sensors are used to measure and report physical movements like acceleration, angular momentum, and force. Our application will collect and analyze data from PPG sensors to detect arrhythmia and pulse frequency to determine if they are within the expected ranges for healthy adults. It will supplement Afib detection with IMU data to determine if a user's movements correlate to those expected when taking prescriptions.

The application will employ a stack of machine learning (ML) algorithms to evaluate data collected by PPG and IMU sensors. Stacking is a method in ML that combines the predictive outputs from multiple base models to determine a final prediction with a greater overall accuracy than the individual models. We will utilize three stacked data collection models to classify PPG and IMU data. Support Vector Machine (SVM), Random Forest (RF), and Multi-layer Perceptrons (MLP) are methods that have been shown to be 97% effective in detecting arrhythmias and abnormal changes in blood pressure when compared to existing FDA approved devices [4]. A differentiating factor of our algorithm will be the type of measurement data that is modeled. Our product uses convolutional neural network (CNN) based wavelet spectrogram images, which are Fourier frequency plots generated by data collected by the PPG. This stack of three base models is trained on existing data consistent with normal sinus rhythms (healthy heart beat), and when applied to a user's CNN wavelet spectrogram images, it can detect the differences attributable to A-fib. An image of the CNN wavelet spectrogram that is fed through our stack to extract key data can be noted below.

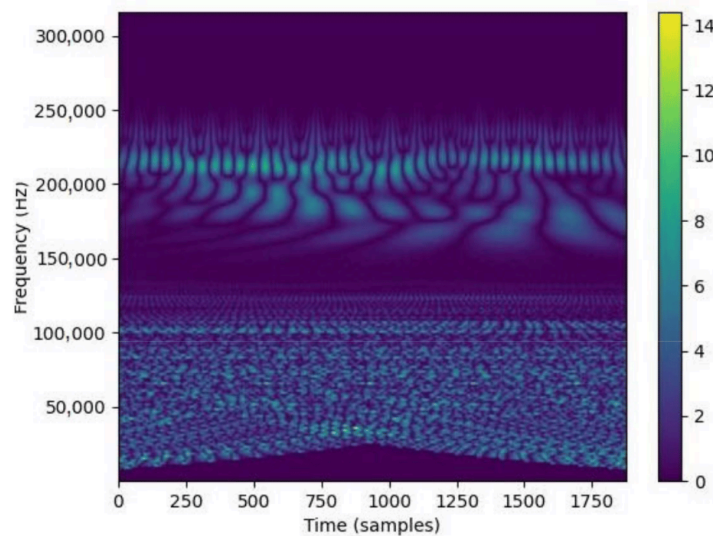


Figure 2 - CNN Wavelets are Fourier transforms of PPG data taken by the device. These are then used to extract key data points like mean, variance, and energy [4].

Part 4: Target Customer: The Evidence

As of 2021 [5], the global market for medical health apps was estimated to be valued at US\$36.1 billion. Growth in this sector is projected at a 14.3% annualized rate, meaning there is a financial opportunity of US\$110.7 billion over the decade spanning 2022-2032 [5]. Our application can be considered a diagnostic app, whose sector is estimated to grow at an annualized rate of 15.9%, thus outperforming the broad medical app market between 2022 and 2032. Surveys of smartwatch users indicate that 82% use fitness and health tracking features, and 79% use notifications and alerts [6]. Statista and the Pew Research Center report that 15% of smartwatch users are adults aged 55 and up, and are specifically motivated to use technology to track and monitor health. This provides an opportunity for market penetration in this growing sector.

Our product, HeartWatch, aims to provide comprehensive support with personalized health monitoring and medication management. It will focus on a variety of widely recognized risk factors among AFib patients for medication non-adherence: physical inactivity, cognitive decline, limited health literacy, and poor mental health. Our application in such cases would help the user to take their medication on time, provide continuous monitoring of the patient's heart, and get them notified when any of the vital signs are abnormal. These features will enable users to take immediate actions and hence consult the doctor at once, and thus prevent emergency conditions like stroke and heart failure.

Part 5: Competition: The Evidence

The two biggest competitors to the HeartWatch app are the Apple Medications feature in the Health app and the Medisafe app. Both Apple Health and Medisafe allow for multiple medications and their respective strengths to be entered. A custom schedule can be created that allows for notifications and critical reminders when medications are forgotten, as well as caregivers to be notified in case of forgetting to take a dose [7]. The apps also alert whether there is a harmful drug interaction when two medications are taken together. The Medisafe app has one additional feature that Apple Health is lacking, which is the ability to add symptoms and track them [8].

The main drawback of both these apps, however, is that they do not implement any health data tracking, such as heart rate or blood oxygen levels. The HeartWatch app asks for the user's permission to collect health data from an external wearable, such as an Apple Watch and FitBit, and uses this data to track whether the medication is causing the user's vitals to act in an expected way. This is an essential benefit of the HeartWatch app compared to the other two, because it allows caregivers to immediately know if the medication prescribed isn't working as expected and a change is needed.

Another issue with the Apple Health app is that it is only available on the iPhone, iPad, and Apple Watch. In the United States, 53.1% of people currently own iPhones [9]. While this is the majority of the country, there is still a significant portion of the population that is not using Apple devices. The HeartWatch app is available on both the Apple app store, the Google Play store, and on the HeartWatch website for direct download without the need of going through a specific store.

Part 6. Method or Approach: The Evidence

The total time expected to bring this app to market is expected to take 35 weeks [10]. As seen in Table 1, app design and feature development will occur simultaneously, with feature development expected to take ten extra weeks. After the groundwork of features for our app has been developed, testing and iterative app updates will follow for a period of 15 weeks. The app is then expected to be released during week 35.

Table 1: Timeline	Weeks 1-5	Weeks 6-10	Weeks 11-15	Weeks 16-20	Weeks 21-25	Weeks 26-30	Weeks 31-35
App Design							
Backend/Feature Development							
Testing							
Iterative Updates							
Finalize and Release							

Table 1: Timeline of complete app development and release.

The total cost to bring this app to market is \$245,385. The biggest cost associated with bringing this project to market stems from employee salaries. Four total app developers will be hired with one focused on design, another on feature development, and two on quality assurance. The average yearly salary for one app developer is \$132,000 and \$98,500 for a project manager [11,12]. Table 2 breaks down these salaries into a dollar-per-week basis and multiplies this rate by the contract length for each employee based on the Table 1 timeline. All these totals are then added up and equate to a total of \$243,985 for labor costs.

Labor Type	Weekly Rate (\$/week)	Duration (weeks)	Total
Design Interface Developer	\$2538	15	\$38,070
App Feature Developer	\$2538	25	\$63,450
Quality Assurance 1	\$2538	15	\$38,070
Quality Assurance 2	\$2538	15	\$38,070
Project Manager	\$1895	35	\$66,325
Total			\$243,985

Table 2: Total labor costs with breakdown.

There are costs other than labor that also need to be considered [13]. Privacy practices that allow for compliance with the Health Insurance Portability and Accountability Act (HIPAA) amount to a total cost of \$1,040 [14]. For the app to run, it will also need to be hosted by a server such as Amazon Web Services [15]. The cost associated with server hosting for the duration of development is \$360, with a detailed breakdown available in Table 3.

Cost Source	Rate	Total
HIPAA	\$1,040/yr	\$1,040
Server Host	\$40/month	\$360
		\$1,400

Table 3: Breakdown of the cost of services.

Part 7: Risk Management: The Evidence

The most important issue in our application is the potential for undetected misuse, especially forgetting to wear the device and technology unfamiliarity among elders. To address this, the application will include optional device-wear reminders and notifications to related caregivers if multiple alerts go unanswered. Besides, recent studies indicate that today's older adults are becoming increasingly familiar with technology, which may enhance their engagement with such features and improve adherence [17].

Overdependence could be another issue. Although current machine learning algorithms can achieve extremely high detection rates (more than 99% according to recent models), it is still essential for users to stay aware about their own symptoms. If users experience any symptoms without receiving any alert from the app, they still need to contact their healthcare provider or visit a hospital to prevent potential emergencies. Although our app aims to reduce unnecessary hospital visits, it is still encouraged for users to maintain regular health check-ups, even if no notifications occur.

The last concern is data security, as some people today underestimate the importance of privacy, not only among elders [18]. This ignorance happens not only in our health monitoring app, but also the other aspects in healthcare systems as we are living in the digital era. In order to protect the sensitive health information, our app will incorporate strong password protection and provide clear guidelines on maintaining data privacy. Additionally, we aim to educate users about the importance of data security through tutorials and periodic updates, using simple explanations and possibly include practical examples.

Part 8: Conclusion

Our application, HeartWatch, presents a practical solution for AFib patients, especially among elderly users, by integrating health monitoring with medication adherence. It can also let users manage their conditions with much more confidence, using wearable technology while putting data security at the forefront. As technology becomes ever more integral to healthcare, HeartWatch will contribute to both individual well-being and efficiency in overall healthcare.

Part 9: Reference

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